Notes of Model Identification and Data Analysis

A series of notes on the course MIDA modulo 1 as taught in Politecnico di Milano by Sergio Bittanti during the academic year 2018/2019

Notes of Model Identification and Data Analysis
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  - Explain the Bayes problem and discuss the Bayes formula
  - In HTE prediction theory for stationary processes in external representation form i.e. input/output models, the canonical representation has a main role, explain what it is meant by canonical representation specifying what is the input signal and what is the output signal of the representation and then highlight its properties. what is the role of this properties in the prediction theory?
  - Briefly explain the concept of minimum variance control for a given plant described by an ARX model, when can this method be used and when not?
  - How can we estimate both the state x(t) and the unknown parameter a of the system $\begin{cases} x(t+1) = ax(t) + v_1(t) \\ y(t) = x(t) + v_2(t) \end{cases}$ by means of the extended Kalman filter
  - A measure problem in model identification is the choice of the model type and the selection of its appropriate complexity. Concisely present the available methods.

Side Notes

Childhood Memories

- $ae^{jw} = acos(w) + jasin(w)$
- $|z_1|^2 = z_1 \bar{z}_1$
- $e^{j\omega} + e^{j(-\omega)} = 2cos(\omega)$
- ***Gain of a transfer function***
  if it's mathematically doable, $\mu = G(s=0) = G(jw=0)$, so in model analysis $\mu = G(z=1)$ since $z = e^{jw}$
- ***From the algebraic form to the exponential one***
  $z = a + jb = re^{jw}$

$$r = \sqrt{a^2 + b^2}$$

$$\theta \in (-\pi, \pi] = \begin{cases} \frac{\pi}{2} & se\ a = 0, b > 0 \\ -\frac{\pi}{2} & se\ a = 0, b < 0 \\ non\ definito & se\ a = 0, b = 0 \\ \arctan(\frac{b}{a}) & se\ a > 0, b \geq 0 \\ \arctan(\frac{b}{a}) + \pi & se\ a < 0, b \geq 0 \\ \arctan(\frac{b}{a}) - \pi & se\ a < 0, b < 0 \end{cases}$$

$$\theta \in [0, 2\pi) = \begin{cases} \frac{\pi}{2} & se\ a = 0, b > 0 \\ \frac{3\pi}{2} & se\ a = 0, b < 0 \\ non\ definito & se\ a = 0, b = 0 \\ \arctan(\frac{b}{a}) & se\ a > 0, b \geq 0 \\ \arctan(\frac{b}{a}) + 2\pi & se\ a > 0, b < 0 \\ \arctan(\frac{b}{a}) + \pi & se\ a < 0, b\ qualsiasi \end{cases}$$

- ***Frequency Response Theorem***
  Let $u(t) = A\sin(\bar{w}t + \varphi)$. If the system is asymptotically stable, at the end of the initial transitory, the output is sinusoidal as well.

$$y(t) = B\sin(\bar{w}t + \theta)$$

$$\begin{cases} B = A|G(j\bar{\omega})| \\ \theta = \varphi + \angle G j\bar{\omega} \end{cases}$$

Observations

- in an $MA(n)$ all the covariances $\gamma(\tau)$ with $\tau > n$ are equal to zero: $\gamma(tau > n) = 0$
- A process is stationary if its transfer function is stable (poles and zeros $\leq |1|$) and it's fed by a stationary signal (i.e., a white noise).
  NB: if the transfer function is in canonical form, it is surely stable.
- $2\eta(t)\ \ \eta \sim WN(0, \lambda^2) \rightarrow \tilde{\eta} \sim WN(0, \lambda^2 2^2)$

- In case the transfer function is fed by a white noise with $E[\eta(t)] = 0$ we can simplify the expression in: $\gamma(y(t)) = E[\ (\ y(t)\ )^2\ ]$

Doubts

- is c_0 in MA processes always equal to 1?
- how to find out if a closed-loop system is stable?
- why the results of the predictor found via PEM or via long division are different? when should I use one and why should I use the other? (page 28 exams pdf)
- if a signal has a zero, does its spectrum surely goes to zero at certain frequencies ? (page 33 exams pdf)

Canonical Form

A stochastic process is in canonical form if its transfer function has the following properties:

- $N(z)$ and $D(z)$ have the same order
- $N(z)$ and $D(z)$ are monic (the coefficient of the highest degree variable is 1)
- $N(z)$ and $D(z)$ are coprime (no roots in common)
- zeros and poles are inside the unitary circle ($|z| \leq 1$ and $|p| \leq 1$)

Exercises on Covariance & Spectra

Variance and Spectra Properties

If two signals α and β are independent and $\varphi = \alpha + \gamma$:

- $\gamma_\varphi = \gamma_\alpha + \gamma_\beta$
- $\Gamma(\varphi(t)) = \Gamma(\alpha(t)) + \Gamma(\beta(t))$

All Pass Filter

$y(t) = \frac{1+az}{1+\frac{1}{a}z}\varphi(t) = (\frac{1}{a}\frac{1+az}{1+\frac{1}{a}z})a\varphi(t) = a\varphi(t)$

Covariance

General method to compute variance and covariance of a process

Transform your process in time domain:
$y(t) = a_1 y(t-1) + a_2 y(t-2) + c_o\eta(t) + c_1\eta(t-1)$

If we recognize that it is an $MA(n)$, we can use a shortcut. We wrote it somewhere, go find it.

Compute $E\big[y(t)\big]$. if it is different from zero, transform it into an equivalent process with expected value equal to zero. {TODO}

Since $E\big[y(t)\big] = 0$ we see that
$\gamma_y(0) = E\big[y(t)^2\big] - E\big[y(t)\big]^2 = E\big[y(t)^2\big]$

$$\gamma_y(k) = E\Bigg[\Big(y(t) - E[y(t)]\Big)\Big(y(t-k) - E[y(t-k)]\Big)\Bigg] = E\Big[y(t)y(t-k)\Big]$$

In order to compute the variance of y do the following:

$$\gamma_y(0) = E[y(t)y(t)] = E\Bigg[y(t)\Big(a_1 y(t-1) + a_2 y(t-2) + c_o\eta(t) + c_1\eta(t-1)\Big)\Bigg]$$

now multiply all the members with one another and take in mind that:

- $E\Big[y(t-k)\eta(t-j)\Big] = 0$ if $k > j$
- $E\Big[y(t-k)\eta(t-k)\Big] = E\Big[y(t)\eta(t)\Big] = E\Bigg[\Big(a_1 y(t-1) + a_2 y(t-2) + c_o\eta(t) + c_1\eta(t-1)\Big)\eta(t)\Bigg]$
- Once you compute $\gamma_y(0)$ or $\gamma_y(1)$ (or even other covariances...) you could obtain values dependent from other covariances, for example $\gamma_y(0) = 2\gamma_y(1)$, but don't worry, in the end you'll have a system of n unknown covariances and n equations.

Compute the Variance of an AR(1) process $y(z) = \frac{1}{A(z)}\eta(z)$

1. Transform the function in the domain of time

2. knowing that $\gamma_{y(t)}(0) = \gamma_{y(t-1)}(0)$ and that $y(t) = y(t-1) + \eta(t)$ then $\gamma_y(0) = \gamma_y(0) + \gamma_\eta(0)$ if and only if y and η are independent, we can solve the last equation because usually the variance of η is known. Another important property of the variance is that $\gamma(ay(t)) = a^2\gamma(y(t))$, given that a is a constant.

Draw the diagram of the prediction error variance as a function of the prediction horizon k

If they ask you such question, probably they already asked you to do the long division for k steps.

Well, use the information gathered from that task, you'll use $Q_1, Q_2, \ldots, Q_k$.

keep in mind that

$$\varepsilon(t|t-k) = Q_k(z)\tilde{e}(t)$$

where $\tilde{e}(t)$ is the WN of the process in canonical form.

Q_k are all known, so just propagate the variance within the equation: $Var(\varepsilon(t|t-k)) = Var(Q_k\tilde{e}(t))$

To do the diagram just put the k in the x axis, and $variance(\varepsilon)$ in the y axis.

Spectra

Compute the spectrum via definition

$$\Gamma(\omega) = \sum_{t=-\tau}^{+\tau} \gamma(t)e^{-j\omega t}$$

Compute the spectrum via magic formula

$$\Gamma(\omega) = |W(e^{jw})|^2 var(\eta(t))$$

Exercise on Minimum Variance Controller

(This is probably just for MIDA2)

First step: compute the k-steps predictor

1. Make sure to be in canonical form.
2. Compute the long division of the transfer function of η, which is $\frac{C(z)}{A(z)}$. we stop at the k-th iteration and we obtain R_k and Q_k (sometimes called E_k).
3. the k-steps predictor is:
   $\hat{y}(t|t-k) = \frac{\hat{R}_k}{C(z)} y(t-k) + \frac{B(z)Q_k}{C(z)} u(t-k)$

Knowing the predictor $y(t+2|t)$, design the Minimum Variance Controller for the system, considering a reference signal $y^0(t)$.

$$y(t) = \frac{C(z)}{A(z)}\eta(t) + \frac{B(z)}{A(z)}u(t-k)$$

We set $\hat{y}(t+2|t) = y^0(t)$.
We are able to set it this way if and only if $B(z)$ has $|z| < 1$.

Isolate $u(t)$.

Draw the block diagram of the control system X
the input is $y^0(t)$, the output is $y(t)$. take the expression of y that is provided to you by the text.

Optimal Estimates of Parameters via PEM

Tips

- sometimes, in order to solve the systems it's handy to compute, for example, $(c_1 + c_2)^2 = \frac{\gamma(0)+2\gamma(1)}{var(e(t))}$ and $(c_1 - c_2)^2 = \frac{\gamma(0)-2\gamma(1)}{var(e(t))}$.

For MA Processes

Suppose to have infinitely many data generated by the following system:

$$\delta : y(t) = 2\eta(t) + 4\eta(n-2)$$

where $\eta(t) \sim WN(0.1)$

For such data, consider the model classes:

$$M_1(a) : y(t) = a_1 y(t-1) + \xi_1(t)$$

$$M_2(a) : y(t) = a_1 y(t-2) + \xi_2(t)$$

where $\xi_1(t) \sim WN(0, \lambda_1^2)\ and\ \xi_2(t) \sim wn(0, \lambda_2^2)$

By using the prediction error minimization method, find the optimal estimates of parameters a_1 and a_2.

in this case we have a $MA(2)$ for δ.

This means that we can compute the covariances in the following way:
Knowing that $y(t) = c_0\eta(t) + c_1\eta(t-1) + \ldots$

$$\gamma(0) = (c_0^2 + c_1^2 + \ldots c_{n-1}^2)\gamma(\varepsilon(t))$$

$$\gamma(1) = (c_0c_1 + c_1c_2 + \ldots + c_{n-2}c_{n-1})\gamma(\varepsilon(t))$$

$$\gamma(2) = (c_0c_2 + c_1c_3 + \ldots + c_{n-3}c_{n-1})\gamma(\varepsilon(t))$$

$$\gamma(3) = (c_0c_3 + c_1c_4 + \ldots + c_{n-4}c_{n-1})\gamma(\varepsilon(t))$$

Let's consider only the first model:

$$M_1(a) : y(t) = a_1 y(t-1) + \xi_1(t)$$

we want it in the form:

$$y(t) = \frac{C(z)}{A(z)}\xi_1(t)$$

and we obtain:

$$C(z) = 1; \quad A(z) = 1 - a_1 z^{-1}$$

we can now write the predictor of y:

$$\hat{y}(t|t-1) = \frac{C(z) - A(z)}{C(z)} y(t) = a_1 z^{-1} y(t)$$

And thus compute the prediction error ε:

$$\varepsilon(t) = y(t) - \hat{y}(t|t-1) = y(t) - a_1 z^{-1} y(t)$$

Now we can compute J, the performance index based on the prediction error:

$$J = E[(\varepsilon(t))^2] = E[(y(t) - a_1 y(t-1))^2] = E[y^2(t)] + a_1 E[y^2(t-1)] - 2a_1 E[y(t)y(t-1)]$$

We know that, being $\eta \sim WN(0,1)$, we have $E[y^2(t)] = \gamma_y(0) = E[y(t-1)]$

so:

$$J = \gamma_y(0) + a_1^2 \gamma_y(0) - 2a_1 \gamma_y(1)$$

and we derive J wrt a_1 and equal it to zero

$$\frac{\delta J}{\delta a_1} = \ldots = 2(-\gamma_y(1) + a_1 \gamma_y(0)) = 0$$

Thus obtaining

$$\hat{a}_1 = 0$$

We repeat the process to find a_2 using the model M_2

For AR Processes

Knowing that $\gamma_v(0) = 10, \gamma_v(\pm 1) = 0, \gamma_v(\pm 2) = 0, \gamma_v(\pm 3) = 3, \gamma_v(\tau) = 0 \; \forall \tau : |\tau| \geq 4$.
Consider the model class

$$M : v(t) = av(t-1) + bv(t-2) + cv(t-3) + \eta(t)$$

with $\eta \sim WN(0, \lambda^2)$

By using the prediction error minimization method, find the optimal estimates of the parameters a, b and c.

The predictor is equal to the starting equation but for the η component:

$$\hat{v}(t|t-1) = av(t-1) + bv(t-2) + cv(t-3)$$

The prediction error ε is always:

$$\varepsilon(t) = v(t) - \hat{v}(t|t-1)$$

Now let's compute J:

$$J = E[(\varepsilon(t))^2] =$$

$$E[v^2(t) + a^2v^2(t-1) + b^2v^2(t-1) + c^2v^2(t-3) +$$

$$-2av(t)v(t-1) - 2bv(t)v(t-2) - 2cv(t)v(t-3) +$$

$$2abv(t-1)v(t-2) + 2acv(t-1)v(t-3) + 2bcv(t-2)v(t-3)] =$$

$$\gamma_v(0) + a^2\gamma_v(0) + b^2\gamma_v(0) + c^2\gamma_v(0) - 2a\gamma_v(1) - 2b\gamma_v(2) - 2c\gamma_v(3) + 2ab\gamma_v(1) + 2ac\gamma_v(2) + 2bc\gamma_v(1)$$

$$J = \gamma_v(0)(1 + a^2 + b^2 + c^2) - 2c\gamma_v(3)$$

To minimize J we observe that we should impose $\hat{a} = \hat{b} = 0$

and, as always, let's derive wrt c:

$$\frac{\delta J}{\delta c} = 2c\gamma_v(0) - 2\gamma_v(3) \rightarrow \hat{c} = \frac{\gamma_v(3)}{\gamma_v(0)} = \frac{3}{10}$$

Parameter Estimation

Knowing Variance

MA

Usually asks which is the most suitable model class for representing the available data.
If you notice that $\gamma(\tau) = 0$ for $\tau > n$, where n is a finite number, you can say that the most suitable model class is a $MA(n)$.

Write the general equation for MA processes:

$$v(t) = e(t) + a_1 e(t-1) + a_2 e(t-2) + \ldots a_n e(t-n)$$

and use the known formulas for the covariances of MA processes to build a system of $n+1$ equations, then solve it.

Knowing Measurements

2. The measurements $y(t)$ of a time series for $t = -1,0,1,2,3,4$ are given in the table below.

t	-1	0	1	2	3	4
$y(t)$	0	1	-1	2	0	1

The aim is to identify, from the available data, a black-box model of the following class

$$m: y(t) = ay(t-1) + by(t-2) + e(t)$$

where $e(t) \approx WN(0, \lambda^2)$.

2.1. Compute the optimal estimate of a, b, and λ in such a way that the following cost function is minimized

$$J = \frac{1}{4}\sum_{t=1}^{4}(y(t) - \hat{y}(t/t-1))^2$$

dove $\hat{y}(t/t-1)$ optimal one-step predictor.

$A(z) = 1 - az^{-1} - bz^{-1}$ $C(z) = 1$

$\hat{y}(t|t-1) = a\,y(t-1) + b\,y(t-2)$

Therefore, defining $\varepsilon(t) = -\hat{y}(t|t-1) + y(t)$,

$\varepsilon(1) = -1 - a$
$\varepsilon(2) = 2 + a - b$
$\varepsilon(3) = b - 2a$
$\varepsilon(4) = 1 - 2b$

$\Rightarrow J = \frac{1}{4}\left\{(-1-a)^2 + (2+a-b)^2 + (b-2a)^2 + (1-2b)^2\right\}$

$$\begin{cases} \frac{\partial J}{\partial a} = 0 \\ \frac{\partial J}{\partial b} = 0 \end{cases}$$

WE OBTAIN $\Rightarrow$ THE SYSTEM:

$$\begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -3 \\ 4 \end{bmatrix} \Rightarrow \begin{matrix} \hat{a} = -\frac{2}{9} \\ \hat{b} = \frac{5}{9} \end{matrix}$$

$\hat{\lambda} = J(\hat{a}, \hat{b})$

Optimal Predictors via Long Division (only $e(t)$ case)

- take the process equation and put it in canonical form
- once you do it you'll have something in the form $y(t) = \frac{C(z)}{A(z)} e(t)$
- $C(z)$ on the left, $A(z)$ on the right. Do the long division:
  Let's for example compute the optimal 1-step predictor:
  Do the long division between $C(z)$ and $A(z)$. you'll obtain a quotient Q_1 and a remainder R_1, and such remainder can be called the 1-step remainder. R_1 and Q_1 will be used for the 1-step predictor in the following way:

  $$\hat{y}(t|t-1) = \frac{\hat{R}_1}{C(z)} y(t)$$

  where $\hat{R}_1$ is R_1 considering only the coefficients of z.
  So, if you want to compute the k-step predictor, just compute Q_K and R_k (which are the results of the k-th iteration of the long division) and use the generalize formula:

  $$\hat{y}(t|t-k) = \frac{\hat{R}_k}{C(z)} y(t)$$

  take care, if a z term is missing, for example $y(t) = 1 - z^{-2}$, the predictor of the missing z term is the same of the predictor of the next z term's corresponding step. so in this case the 1-step predictor is the same of the 2-steps predictor.

Questions

explain what is the purpose of the final prediction error FPE method and how it can be used in practice given a set of data

Explain the Bayes problem and discuss the Bayes formula

In HTE prediction theory for stationary processes in external representation form i.e. input/output models, the canonical representation has a main role, explain what it is meant by canonical representation specifying what is the input signal and what is the output signal of the representation and then highlight its properties. what is the role of this properties in the prediction theory?

Briefly explain the concept of minimum variance control for a given plant described by an ARX model, when can this method be used and when not?

How can we estimate both the state x(t) and the unknown parameter a of the system $\begin{cases} x(t+1) = ax(t) + v_1(t) \\ y(t) = x(t) + v_2(t) \end{cases}$ **by means of the extended Kalman filter**

A measure problem in model identification is the choice of the model type and the selection of its appropriate complexity. Concisely present the available methods.